# ANALYSIS OF OPTIMAL MAINTENANCE INTERVAL ON ID FAN USING RELIABILITY CENTERED MAINTENANCE

Muhammad Fakhrul Muqauwim
Power Plant Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
owim@pg.student.pens.ac.id

Hendrik Elvian Gayuh Prasetya
Power Plant Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
hendrik@pens.ac.id

Radina Anggun Nurisma
Power Plant Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
dina@pens.ac.id

***Abstract*— Induced draft fan plays an important role to pull out the flue gas. However, there are some failures regarding ID fan operation. In this case, it is found out that the industry has a long maintenance interval to overcome the failures. This research is conducted to search for maintenance intervals optimally for each failure mode using RCM (Reliability Centre Maintenance) method. To maintain the reliability of the ID fan remains in the expected value, it needs appropriate maintenance strategies. The results obtained on the fan bearing with the failure mode, damage to the bearing (resulting in high vibration) has a maintenance interval every 79.582 h, damage to the bearing (resulting in high temperature) has an interval every 35.887 h, unbalance has a maintenance interval every 46 hours, misalignment has a maintenance interval every 46.9575 hours. While on the impeller with the failure mode, dirty impeller has a maintenance interval every 97.1575 hours. On the motor bearing with the failure mode, in the form of quality and quantity of motor bearing grease, that has maintenance interval every 66.8465 hours. Then, for the inlet guide vane bearing with the failure mode, jammed/damaged bearing has a maintenance interval every 61.79 hours. Thus, this proposed maintenance interval has a shorter interval than existing interval in power plant industry. With shorter maintenance interval, hope it can decrease the failures in the following day, so the industry will get fewer losses due to the failures.**

***Keywords*— *ID Fan, Reliability, Maintenance, Interval***

## I. INTRODUCTION

ID fan is an equipment that can provide a negative pressure in the boiler so that the circulation of the exhaust gas goes well. However, it is often on the ID fan, there are some problems in the operation there. Things like this can certainly shorten the life and affect the performance of the ID fan. EPRI, describes over 50% of the axial ID fan problems were with blades, shaft bearings, and blade thrust bearings [1].

The optimal maintenance strategy is needed in maintaining the performance of the ID fan. One of maintenance method that can be applied in maintaining the performance of the ID fan remains on the level of reliability desired is Reliability Centered Maintenance. Reliability Centered Maintenance (RCM) is the maintenance technique that maximizes reliability at lowest possible cost [2]. Besides, RCM is also a method of maintenance that has been successfully applied in some countries to get the profit in the economy [3]. In addition to some superiority in terms of costs of maintenance, RCM also has the superiority to determine the maintenance activities that are applicable and effective to keep the reliability of the system and the equipment [4].

In its application, RCM uses two types of methods, namely qualitative and quantitative. Qualitative methods using the application of FMEA and RCM II Decision Diagram. FMEA (Failure Mode and Effect Analysis) is a method used to identify the kind of failure that might cause any failure and to ensure the influence of failures associated with each form of failure [4]. While the RCM II Decision Diagram is a method that has advantages in the determination of the maintenance program that is focused on the critical components and avoids maintenance activities that are not required to specify the interval for its proper maintenance. The expected target on the qualitative method was able to analyze the layout of the critical components and determine the type of treatment that will be applied.

Quantitative methods on the RCM is a development of the last RCM [5]. While in the quantitative method the calculation is done to get the value of the maintenance interval. On the maintenance interval, is used the value of MTTF, MTTR, replacement cost, worker cost, production losses, and the cost of downtime. However, the use of the data has been mentioned adjusts with the maintenance task selected.

Reference [6] added FTA (Fault Tree Analysis) to determine root causes of failures that occur on the mill and using the Minitab 16 for testing the distribution with the terms of the hypothesis are accepted if P-Value > $\alpha$, with $\alpha$=0,05. N. Sembiring and G. A. Elvira added LTA (Logic Tree Analysis) to analyze the critical part at the screw press and using the Minitab 16 to select the distribution with the largest correlation of coefficient [7]. While Arif, Juniani, and Indrawan only using FMEA and RCM II decision diagram to analyze failure in flame plasma cutting using Weibull 6++ to determine the distribution of the MTTF and MTTR calculation [8].

To achieve the level of reliability desired and optimal lifetime obviously requires optimal maintenance activities. However, often maintenance activities are applied without any scheduling and do not have a strong reason for the determination of maintenance intervals. Therefore, in this study, it is expected to get the optimal maintenance interval, to support the performance of the ID fan remains optimal.

## II. ID FAN PRINCIPAL AND COMPONENTS

### *A. ID Fan Principal*

The ID fan has the function as the prime mover exhaust gases from the furnace towards the stack. Also, the ID fan serves as a control pressure and flow in the furnace. As in Fig. 1, it starts from the inflow of the exhaust gas through the suction casing which further flows through the inlet guide vane by using the suction power of the impeller before finally flowing into the damper and is discharged to the stack.

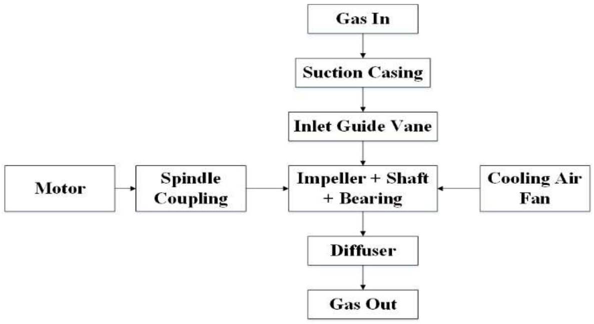


**Fig. 1. ID fan functional block diagram**

The ID fan (that studied in this research) using axial flow fan type. In axial flow fan, flow and pressure can be controlled using variable-pitch blades, inlet vanes, or variable-speed drives [1]. The type of control namely inlet guide vane. In this type, the blade does not rotate. On the IGV there is a vane that can be set as an opening angle.

### *B. ID Fan Components*

On the ID fan, there are several components, including:

1) Suction casing serves as an air inlet also as a cover. At this stage, the air enters with the angle of entry by 90 degrees

2) Motor serves as the driving force of the impeller. The age of the motor depends on the layer of insulation. And also depends on the amount of use and operating temperature. [1].

3) Spindle coupling used to continue torque in large quantities. Spindle coupling has 2 parts, namely transmission long shaft and transmission short shaft [1].

4) The inlet guide vane serves as a regulator of the furnace pressure and regulating the amount of flow that will be discharged to the stack. The type used is an adjustable static blade. It means the IGV are not turning but the angle of the vane can be set. If the pressure in the furnace is less negative, the opening angle vane must be opened more.

5) The shaft serves as the successor of the motor rotation to the impeller. The rotating force in the shaft axial fan is low and the bearings are located close to the rotor. Shaft in the axial fan has a small diameter [1].

6) The bearing serves as a support for shaft and impeller. In the axial flow fan, most use bearing types are ball and roller bearings. This type has four main components; outer race, inner race, rolling elements, and spacers of the rolling elements [1].

7) The impeller serves as a producer of negative pressure on the suction side as well as serves to generate positive pressure on the discharge side so the flow of boiler flue gas can flow into the stack. In the event of an extended shutdown, the impeller should spin a few turns to ensure the lubrication at the bearing in good condition [1].

8) The cooling fan used to provide the clean and cool air, to cool the main shaft bearing [1].

9) The diffuser is located after the impeller for directing the gas that will be discharged through the stack.

10) The damper serves as an input channel of the exhaust gas when the ID fan is operating and as fan isolation when doing maintenance.

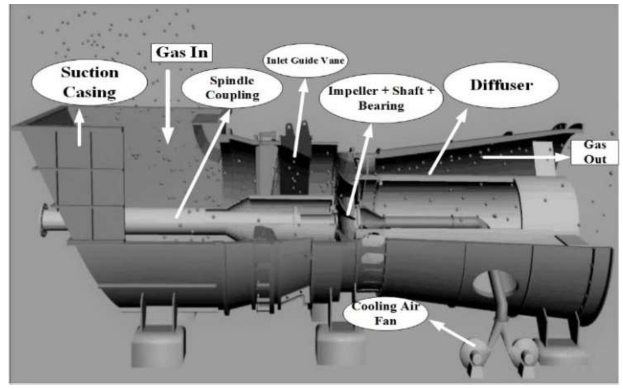


**Fig. 2. ID fan section view**

## III. RELIABILITY CENTRE MAINTENANCE II

### *A. Maintenance*

Maintenance actions are divided into corrective maintenance (CM) and preventive maintenance (PM) [9]. CM actions are unscheduled and are intended to restore a system from a failed state to a working state through either repair or replacement of failed components [9]. On the other hand, PM is scheduled and can be carried out either to reduce the failure or to improve the availability of the system [9]

Maintenance has direct and indirect impacts on sustainability-related aspects. Directly, through the execution of maintenance activities, and indirectly, on production processes and on final product quality due to maintenance efficiency/inefficiency [10]. Meanwhile, the goal of maintenance activities is to make sure that the physical assets can able to operates as the user desired in its function [11].

### *B. Downtime*

Downtime appears when a machine stops until the machine is restarted [12]. The longer the downtime of equipment, it will be more losses incurred. Therefore, the proposed maintenance task expected to minimize the value of downtime.

### *C. FMEA (*Failure Modes and Effect Analysis*)*

FMEA is an effective tool for identifying and eliminating risk in many fields, so it is necessary to rank each failure mode accurately [13]. Failure Mode and Effect Analysis (FMEA) is classified as the semi-quantitative method. FMEA uses RPN (Risk Priority Number) to determine the impact of risk (severity), the possibility of risk (occurrence), and risk opportunities (detection). The value of RPN is obtained by [14]:

$$S \times O \times D = RPN \quad (1)$$

Where :

S = Severity (level from the impact of risk)
O = Occurance (level from the possibility of risk)
D = Detection (level from the detection of risk)

### *D. Realibility Centered Maintenance II*

Reliability centered maintenance is a process used to determine what should be done to ensure any physical items or a system following the functionality desired by its users [11]. RCM focuses on handling to take care of a component to awake the value of the reliability by considering the cost-effectiveness of maintenance.

### E. MTTF and MTTR

MTTF (Mean Time to Failure) of a system is given as the average time between the failures of a system [15]. MTTR (Mean Time to Repair) is the average time taken by a system to recover from any failure [15]. The equation of MTTF and MTTR on each distribution has a different value. So also the parameters inside. The following equation to calculate the MTTF and MTTR in each type of distribution is:

- Normal Distribution :

The parameters used in normal distribution is μ (the middle value) and σ (standard deviation). The normal distribution is often called Gaussian distribution, where this distribution has a characteristic symmetrical around the average with a spread in the distribution of which is determined by the σ [16].

$$MTTF = \mu \quad (2)$$

$$MTTR = \mu \quad (3)$$

- Lognormal Distribution :

Lognormal distribution using two parameters, namely s as *shape parameter* and $t_{med}$ as *location parameter* which is the median value of the distribution [16].

$$MTTF = t_{med} e^{\frac{s^2}{2}} \quad (4)$$

$$MTTR = t_{med} e^{\frac{s^2}{2}} \quad (5)$$

- Exponential Distribution :

The exponential distribution has a failure rate that is constant against time. This distribution is most easy to analyze. The distribution parameter used is λ (failure rate) [16].

$$MTTF = \frac{1}{\lambda} \quad (6)$$

$$MTTR = \frac{1}{\lambda} \quad (7)$$

- Weibull Distribution :

The Weibull distribution is an empirical distribution that has a θ parameter, which is called the scale parameter, and β is called the shape parameters. The β parameter is useful to determine the level of damage from formed pattern data, and the scale parameter (θ) affect the data median [16]

$$MTTF = \theta\Gamma(1+\frac{1}{\beta}) \quad (8)$$

$$MTTR = \theta\Gamma(1+\frac{1}{\beta}) \quad (9)$$

### F. Optimal Maintenance Interval

The results of the RCM II information worksheet (FMEA) analysis, decision worksheet, and distribution analysis will be used to provide proposed maintenance activities on each failure mode along with its maintenance interval [17]. The formula for calculating maintenance interval for scheduled on condition task, namely [18]:

$$PM = ½ \text{ x P-F Interval} \quad (10)$$

Where :

PM = Maintenance interval for schedule on condition task
P-F Interval = MTTF (*Mean Time to Failure*) [18]

Meanwhile, to calculate the maintenance intervals for the scheduled restoration tasks and scheduled discard tasks should be done the calculation of the repair or damaged components replacement cost [18]. The formula used is :

$$TM = \eta \text{ x } \left(\frac{C_m}{C_f(\beta-1)}\right)^{\frac{1}{\beta}} \quad (11)$$

$$Cf = Cr + MTTR\ (Co + Cw) \quad (12)$$

$$Cm = Downtime\ cost + Cw + Cf \quad (13)$$

Where :

TM = Maintenance interval for scheduled restoration task or scheduled discard task
η = Distribution parameter
Cf = Repair cost
Cr = Replacement cost
Co = Loss of production
Cw = Worker cost
Cm = Maintenance cost

## IV. Results and Discussion

At this stage, results and analysis of the optimal maintenance interval in ID fan with RCM method will be presented.

### A. ID Fan Downtime

Calculation and analysis begin with analyzing the value of the ID fan downtime to know the critical unit at the ID fan. Unit with the highest downtime, it will be the focus of discussion [19].

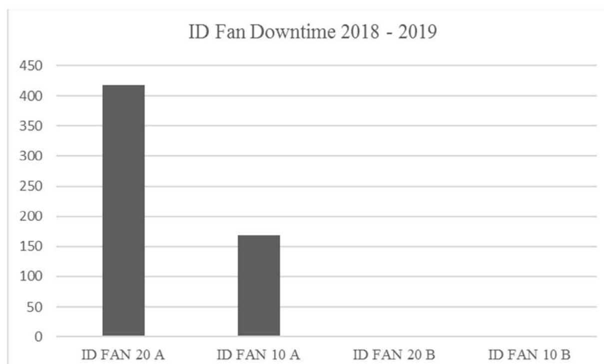


**Fig. 3. ID fan downtime 2018 – 2019**

From Fig. 3, can be known if, ID fan unit 20A has a highest downtime value by 418,18 hour compared to the ID of the fan unit 10A which has a downtime value by 168,12 hour. While on the ID fan 10B and 20B are not encountered downtime. Then, the unit 20A becomes the focus of discussion in this study, due to having the highest downtime.

### B. FMEA

Next is the preparation of the ID fan failure into the FMEA. After that can be done by spreading the questionnaire to get the value of RPN, which will subsequently be used to determine the critical components on the ID fan.

Values of RPN can vary from a maximum of 1000, the greatest risk, to a minimum of 1. Numbers (RPN) derived are often used to select the “vital few” problems to work on. This can be done by setting a threshold limit, for example, RPN 200, and working on all potential failures above this limit [20]. Another approach is to arrange the RPN values in a Pareto plot and give attention to those potential failures with the highest ratings [20]

This research will use the limit of the RPN by 200 for the selection of failure modes to be prioritized. Here is a chart of the overall RPN :

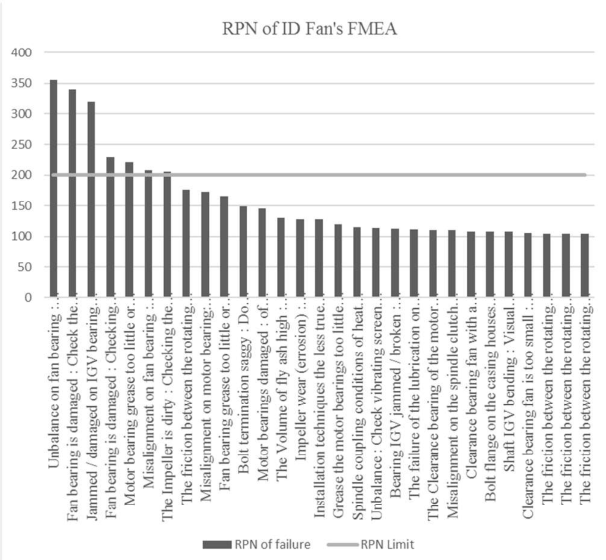


**Fig. 4. RPN of ID Fan's FMEA**

There are some failures on the components which have the RPN value more than 200 based on Fig.4 above. The component that most dominates the RPN value above 200 is fan bearing. Failure modes such as unbalance, misalignment, and damages on fan bearing causing effects that can be detrimental to the power plant exhaust gas system, namely the fan trip.

Besides, some components are prioritized for immediate maintenance, i.e. IGV bearing, motor bearing, and impeller. The failure mode on the IGV bearing is jammed or damaged bearing. The failure mode on the motor bearing is the lack of grease as a lubricant on the bearing. Meanwhile, the failure mode on the impeller is a dirty impeller resulting in vibration which can interfere with the performance of the fan.

### C. *RCM II Decision Worksheet*

Making the RCM II Decision Worksheet by using the critical components of the FMEA processing results. The results obtained can be seen in Table II. RCM II Decision Worksheet can be filled in with the help of the RCM II Decision Diagram.

### D. *MTTF and MTTR value*

On the Minitab 18, after the running determination of the distribution resulting calculation of the MTTF as well. Meanwhile, due to the equation of MTTF = MTTR, then the results of the processed time to repair (TTR) data will also result in the calculation of MTTR. The results of the MTTF and MTTR value can be seen in Table I.

After getting the results of MTTF and MTTR, can be analyzed that the failure mode of bearing damage (with the result of high temperature) has the highest MTTF and MTTR value. This indicates that the failure is most prone to occur, as well as have the longest time to repair.

Meanwhile, failure in the form of a dirty impeller with the result of the unbalance that may cause high vibration, has the lowest MTTF and MTTR value. This indicates the failure mode is rare and have the most quickly time to repair.

TABLE I. MTTF AND MTTR VALUE

| FAILURE MODE | MTTF | MTTR |
|---|---|---|
| Damaged on fan bearing (resulting in high vibration) | 159.164 hour (Weibull) | 24.102 hour (Weibull) |
| Damaged on fan bearing (resulting in high temperature) | 71.774 hour (Normal) | 32.548 hour (Lognormal) |
| Unbalance on fan bearing (resulting in high vibration) | 92 hour (Weibull) | 7.57 hour (Lognormal) |
| Misalignment on fan bearing (resulting in high vibration) | 93.915 hour (Weibull) | 8.237 hour (Lognormal) |
| The Impeller is dirty (resulting in unbalance which can lead to high vibration) | 194.315 hour (Lognormal) | 3.961 hour (Lognormal) |
| The quality and quantity of grease motor bearings low (resulting in high vibration) | 133.693 hour (Weibull) | 13.65 hour (Normal) |
| Jammed/damaged on IGV bearing (resulting in the operation of the ID Fan is interrupted) | 123.58 hour (Lognormal) | 17.037 hour (Normal) |

### E. *Calculation of Optimal Maintenance Interval*

After spreading the RCM Decision Diagram to the expert, finally obtained the maintenance task selected at each prioritized failure. On the prioritized failures are in Scheduled on-Condition Task.

The calculation of the maintenance interval on a scheduled on-condition task is done by calculating ½ of the P-F Interval of each component. P-F Interval used is the data of MTTF of each critical component. The following example of the calculation interval treatments applied to the failure such as misalignment fan bearing :

MTTF = 93.915 hour

PM = ½ P-F Interval = ½ MTTF = ½ 93.915 hour
= 46.96 hour

Through the calculation of the optimal maintenance interval is known if the results obtained are smaller than the MTTF of each failure. This indicates that the need for preventive action before failure occurs, as well as to avoid such a failure occurs then perform the maintenance actions and appropriate monitoring.

TABLE II. RCM II DECISION WORKSHEET

| RCM II DECISION WORKSHEET | | | | System : Flue Gas System | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Sub System : ID Fan | | | | | | | | | | | | |
| ***Component*** | ***Information References*** | | | ***Consequences Evaluation*** | | | | ***H1*** ***S1*** ***O1*** | ***H2*** ***S2*** ***O2*** | ***H3*** ***S3*** ***O3*** | ***Default Action*** | | | ***Proposed Task*** | ***Initial Interval*** | ***Can Be Done by*** |
| | F | FF | FM | H | S | E | O | N1 | N2 | N3 | H4 | H5 | S4 | | | |
| Impeller | 1 | 1 | 1 | N | N | N | Y | Y | - | - | - | - | - | Scheduled on condition task | 97,16 hour | Maintenance staff |
| Fan bearing | 1 | 1 | 1 | N | N | N | Y | Y | - | - | - | - | - | Scheduled on condition task | 35,89 hour | Maintenance staff |
| | 1 | 2 | 2 | N | N | N | Y | Y | - | - | - | - | - | Scheduled on condition task | 79,58 hour | Maintenance staff |
| | 1 | 2 | 3 | Y | - | - | - | Y | - | - | - | - | - | Scheduled on condition task | 46 hour | Maintenance staff |
| | 1 | 2 | 5 | Y | - | - | - | Y | - | - | - | - | - | Scheduled on condition task | 46,96 hour | Maintenance staff |
| Bearing Motor | 1 | 2 | 2 | Y | - | - | - | Y | - | - | - | - | - | Scheduled on condition task | 66,85 hour | Maintenance staff |
| Inlet Guide Vane | 1 | 1 | 3 | N | N | N | Y | Y | - | - | - | - | - | Scheduled on condition task | 61,79 hour | Maintenance staff |

TABLE III. PROPOSED MAINTENANCE INTEVAL

| **Failure Mode** | **Initial Interval** |
|---|---|
| Damaged on fan bearing (resulting in high vibration) | 79.58 hour |
| Damaged on fan bearing (resulting in high temperature) | 35.89 hour |
| Unbalance on fan bearing (resulting in high vibration) | 46 hour |
| Misalignment on fan bearing (resulting in high vibration) | 46.96 hour |
| The Impeller is dirty (resulting in unbalance which can lead to high vibration) | 97.16 hour |
| The quality and quantity of grease motor bearings low (resulting in high vibration) | 66.85 hour |
| Jammed/damaged on IGV bearing (resulting in the operation of the ID Fan is interrupted) | 61.79 hour |

From Table III, it can be known that the failure with the shortest maintenance interval is damaged on fan bearing that resulting in high temperature. Therefore, monitoring using a thermograph should be ensured to run well to get optimal maintenance. Besides, it can be seen in Table IV that the proposed maintenance interval has a shorter value than the existing industry maintenance intervals. It is expected to reduce the losses that occur due to its failure.

TABLE IV. EXISTING MAINTENANCE INTEVAL

| **Monitoring Type** | **Schedule** |
|---|---|
| Vibration | Every week on Tuesday (Every 168 hour) |
| Temperature | Every first and third week on Tuesday (Every 368 hour) |

The difference in maintenance intervals is one factor of the difference in total cost. The proposed maintenance task has a shorter period than the company does. Even though it would make more extravagant in the expenditure of worker cost, but it is expected to prevent the occurrence of failure may result in large losses. The sooner the failure is known, the sooner preventive tasks it will be applied and it is expected a failure does not occur in the future.

## V. CONCLUSIONS

The goal of this study is to figure out the proposed maintenance interval for critical failure modes in the ID fan. As the results of the analysis above, the conclusions are summarized that the proposed maintenance interval has a shorter interval than the existing interval in the power plant industry. With a shorter maintenance interval, it is known to prevent earlier failures, so the industry will get fewer losses due to failures. Then, reliability values will remind the same at the expected values and it will save the expenses.


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